

Deleverging Developers in China: Real Effects and Spillovers*

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Abstract

This study examines how a policy-induced developer credit crunch affects developers, housing markets, and local government finance. Developers breaching the thresholds set by China’s Three-Red-Lines policy experienced sharp declines in debt growth, land acquisitions, and performance. Privately owned developers suffered larger sales losses and higher default risk than state-owned developers. Using project-level launch and sales data, we show that both supply- and demand-side forces contributed to this ownership gap: policy exposure reduced project launches more among private developers, while homebuyers shifted purchases from private developers toward SOEs. As developers reduced land purchases, local governments increasingly relied on LGFVs to buy land, weakening LGFV liquidity and raising financing costs.

Keywords: Three-Red-Lines, Developer Deleveraging, Land Finance, LGFVs

JEL Classification: G30, G31, G38, R10, R30

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1 Introduction

China is home to the world’s largest residential real estate market.¹ Prior research emphasizes that China’s housing market is distinctive not only in its scale, but also in its deep entanglement with local government incentives and the financial system (Glaeser et al., 2017; Rogoff and Yang, 2021; Xiong, 2023; Song and Xiong, 2018). Yet this scale and institutional structure have also made the sector a major source of risk. These risks are especially concentrated among property developers, whose business model has historically relied on high leverage and high turnover.² This setting raises an important policy question: when policymakers attempt to deleverage highly leveraged real estate developers, how does such intervention affect developers themselves and spill over to the broader housing market and local government finance?³

This paper studies these questions using China’s Three-Red-Lines (TRL) policy, implemented on January 1, 2021. The TRL policy imposed explicit and binding balance-sheet thresholds on real estate developers and tied their debt growth to the number of thresholds breached. We examine how this developer-side credit crunch affects developers’ investment, profitability, and default risk; housing-market supply and demand; and the local government financing vehicles (LGFVs).

We assemble four datasets to trace the effects of the policy from developer balance sheets to land and housing markets and local government finance. We collect from WIND financial information on 160 publicly listed real estate firms, with 80 listed on the China A-share market and 80 on the Hong Kong stock market. Land transaction data for those real estate firms come primarily from China Real Estate Information Corp (CRIC) database, a major digital platform for China’s real estate sector that offers comprehensive coverage and

¹Savills Research estimates that global residential real estate was worth \$286.9 trillion at the end of 2024, of which China accounted for about \$74.6 trillion and the United States about \$51.6 trillion.

²China’s real estate sector has entered a period of severe distress: real estate bond defaults totaled RMB 31.1 billion during 2017–2020, but increased to RMB 111.7 billion during 2021–2023 (Figure 1). China Evergrande Group provides the most prominent example: as Reuters reported, when its liquidity crisis intensified in 2021, the developer had more than US\$300 billion in liabilities and roughly 1,300 projects across more than 280 cities; its bond default raised concerns that developer distress could spill over to housing-market confidence and local government revenues.

³Local governments rely heavily on land-transfer revenues: according to the Ministry of Finance, revenue from the transfer of state-owned land-use rights accounted for about 90 percent of local government-managed funds revenue in 2021.

accurately attributes land acquisitions by subsidiaries to their parent firms. We further obtain data on housing project sales and project opening from the China Real Estate Index System (CREIS) database, which provides detailed sales information at the project-quarter level. Local Government Financing Vehicles (LGFVs) and their subsidiaries (Sub-LGFVs) are identified using WIND and the Industrial and Commercial Enterprises Information System, resulting in a database of 3,419 LGFVs and 41,401 Sub-LGFVs.⁴ Together, these data provide a comprehensive basis for analyzing how the TRL policy reshaped both the housing market and local government financing.

Based on these data, we start with assessing the economic consequences of the TRL policy on real estate developers using a difference-in-differences (DID) design with ordered discrete treatments. The treatment variable represents the number of red lines breached by each firm, as determined by its financial position in 2019, before the policy announcement. First, we find that firms crossing one additional red line experienced a 10.6% decline in interest-bearing debt and reduced their land purchase value by 27%. This policy also led to deteriorating financial performance among line-crossing firms. In particular, we find that violating one more red line reduces developers' sales by 6.6%, lowers ROE by 8.3 percentage points, and increases the probability of default by 10.7 percentage points after the policy's implementation in 2021. The absence of significant pre-trends supports the validity of our identification strategy. Importantly, these effects are not uniform. Following the TRL policy, privately-owned developers experience significantly greater financial distress than state-owned peers with comparable violations, suffering a 12.8% decrease in sales, a 9.9-percentage-point drop in ROE, and a striking 15.9-percentage-point increase in default probability. This disparity suggests that the policy disproportionately impacts private developers, potentially altering the competitive landscape of the real estate sector.

We next turn to housing market outcomes. Using project-quarter sales data from CREIS, we confirm that the TRL policy leads to sizable declines in housing sales among listed developers: each additional red-line violation is associated with roughly 12–17% lower sales volume and value, along with a modest decline in average prices. These effects are highly heterogeneous by ownership: relative to SOEs with comparable TRL exposure, private de-

⁴The methodology for identifying Sub-LGFVs through ownership networks is detailed in Section 3.4.

developers experience substantially larger reductions in units sold, sales area, and sales value, indicating a stronger policy impact on private firms.

We then decompose the sales effects into supply- and demand-side channels. On the supply side, developers that breach more red lines launch fewer projects and bring fewer units and less construction area to market after the policy. Crossing one additional red line is associated with roughly a 26% decline in the number of projects opened for sale, and roughly 34–35% declines in newly supplied units and construction area. This supply retrenchment is significantly stronger for private developers, consistent with tighter financing constraints translating into a larger pullback in new projects launches.

On the demand side, we find little evidence that buyer demand varied with the number of red lines crossed by developers. Within-project estimates show economically small and statistically insignificant effects on sales and prices, suggesting that buyers did not substantially discount or avoid projects by more constrained developers. However, the muted average effects mask a sharp ownership split. Within the same local markets, sales decline for private developers as TRL exposure increases, while sales increase for state-owned developers. This pattern is consistent with buyers responding to the completion risk in China’s pre-sale housing system. Buyers do not only choose price, location, and quality; they also assess whether the developer can complete and deliver the project. State ownership becomes a form of implicit guarantee.

Taken together, the TRL policy generates both a pronounced supply contraction and a demand reallocation across developer types. Private developers cut back new supply more aggressively and simultaneously lose demand to SOE projects, whereas SOEs are relatively insulated and may even gain market share in demand. These two forces jointly help explain the widening POE-SOE gap in sales outcomes and financial distress after the policy.

Lastly, we study how the TRL policy spilled over from developers to local public finance, given that land sales are a key revenue source for local governments. After the policy, land purchases shifted sharply away from listed real estate firms toward government-linked buyers, especially Sub-LGFVs.⁵ This reallocation appears to have absorbed part of the land-

⁵Before 2020, real estate firms consistently dominated the primary land market, purchasing substantially more land than LGFVs and Sub-LGFVs combined. After the TRL policy, listed real estate firms’ land purchases fell from RMB 2.63 trillion in 2020 to RMB 0.74 trillion in 2023, while Sub-LGFV purchases rose

market shortfall created by constrained developers, but much of the acquired land was not subsequently brought to market. Matching land parcels to housing project-level opening-for-sale records, we find a large increase in undeveloped land holdings among government-linked entities, with especially sharp increases for Sub-LGFVs.⁶

Because developers exhibit strong purchase inertia, a city’s exposure to the TRL policy depends importantly on its pre-policy reliance on developers that were later constrained by the new regulation. Motivated by this feature of the data, we construct a city-level treatment intensity measure, defined as the share of pre-policy land value sold during 2017–2019 to developers that later crossed the red lines, and use it to trace the policy’s spillover effects on government-linked buyers. Regression results show that in cities with greater treatment intensity, LGFVs significantly increased their land-market propping activity in the post-policy period. In those cities, LGFVs face higher bond yield spreads and lower current, quick, and cash ratios. Together, these results indicate that the TRL policy shifted land demand toward LGFVs and their subsidiaries, but at the cost of increasing contingent fiscal burdens and heightening implicit debt risks.

Our paper contributes first to the literature on China’s local government finance and the real estate-fiscal nexus. Prior work shows that land finance, LGFVs, and local government debt are central to China’s growth model and financial system (Ru, 2018; Chen, He, and Liu, 2020; Huang, Pagano, and Panizza, 2020; Gao, Ru, and Tang, 2021; Gyourko et al., 2022). Chang, Wang, and Xiong (2026) document how local governments managed land and housing prices amid declining transactions during COVID-19. Our findings complement Chang, Wang, and Xiong (2026) by showing that LGFV land purchases often function as land warehousing rather than productive development. We further show that this form of land-market support imposes financial costs on LGFVs, leading to higher bond yield spreads and weaker short-term liquidity. This is consistent with the view that governments can use quasi-public financing vehicles to support short-run activity while accumulating financial

from RMB 0.32 trillion in 2020 to RMB 0.69 trillion in 2022. By 2022, LGFVs and Sub-LGFVs together purchased more land than listed real estate firms.

⁶We match land parcels to subsequent housing projects using project-level opening-for-sale data and classify parcels not matched to later projects as undeveloped. For Sub-LGFVs, average annual undeveloped land value rose by 148%, from RMB 176.4 billion in 2017–2020 to RMB 437.6 billion in 2021–2023. Their development ratio fell from about 12.0% during 2017–2019 to 8.3% in 2021 and 4.6% in 2023.

stability risks (Acharya, 2011).

Second, we contribute to the literature on China’s real estate sector. While much of the existing literature focuses on the sector’s prolonged boom and its economic implications (e.g., Glaeser et al., 2017; Han, Han, and Zhu, 2018; Deng et al., 2022; Rogoff and Yang, 2024; Chen et al., 2025), few studies evaluate the impact of deleveraging reforms such as the TRL policy (Chen, Du, and Ma, 2024). Our findings fill a gap in the literature by showing how the TRL policy reshaped China’s real estate sector: constrained developers sharply reduced land acquisitions and experienced weaker financial performance, with the effects concentrated among POEs and amplified by a demand reallocation from POEs to SOEs.

Third, we contribute to the broader literature on credit supply and housing markets. Existing studies have emphasized several household- and mortgage-side channels, including mortgage credit expansion and house price cycles (Mian and Sufi, 2014; Favara and Imbs, 2015; Di Maggio and Kermani, 2017; Mian and Sufi, 2022; Adelino, Schoar, and Severino, 2025), securitization and credit supply by intermediaries (Keys et al., 2010; Justiniano, Primiceri, and Tambalotti, 2019), and borrower leverage constraints under macroprudential regulation (Acharya et al., 2022). Our setting highlights a complementary channel: developer-side credit supply.⁷ In markets where developers rely on external finance and presales to acquire land, start construction, and complete housing projects, a credit shock to developers can reduce land demand, contract housing supply, dampen buyer demand by increasing completion risk, and spill over to local public finance. By shifting attention from household borrowing constraints to developers’ finance, our paper shows how a regulatory shock to developer balance sheets propagates through housing markets, and government revenues.

⁷For related works on the supply side of the housing market, see also Glaeser, Gyourko, and Saiz (2008), Saiz (2010), Nathanson and Zwick (2018), Baum-Snow and Han (2024), Gu (2024), and Van Straelen (2024).

2 Institutional Background

2.1 Land, Housing, and Public Finance in China

Real estate plays a central macroeconomic role in China. [Rogoff and Yang \(2021\)](#) estimate that real-estate-related activities account for nearly 30% of GDP when direct construction and extensive upstream and downstream linkages are included. Institutionally, China’s urban land market operates through the transfer of long-term land-use rights, while ultimate ownership of urban land remains with the state ([Liu and Xiong, 2018](#)). Local governments act as the primary—and effectively monopolistic—suppliers of new urban construction land in the primary market. Most residential and commercial parcels are allocated through standardized competitive mechanisms (collectively known as zhao-pai-gua), including auction, bidding, and listing formats, in which the highest bidder acquires use rights for a fixed term ([Chen and Kung, 2019](#)). Beyond revenue generation, land is also managed as a policy instrument: zoning restrictions and differentiated pricing across uses are used to pursue multiple objectives ([He et al., 2025](#)). Figure 3 summarizes the institutional chain and financial flows linking local governments, LGFVs, developers, and the housing market.

Land-transfer revenues are the cornerstone of local public finance, and this dependence is rooted in the 1994 Tax-Sharing Reform. This reform recentralized tax revenues while leaving local governments with disproportionate expenditure responsibilities. To bridge this fiscal gap, local authorities turned to quasi-fiscal funding sources, most notably land-based finance. In 2020, gross land-transfer revenues reached RMB 10.01 trillion, equivalent to 84% of local governments’ general budgetary revenue ([Gyourko et al., 2022](#)). Land-transfer revenues fall under the “Government Fund Budget.” They are closely linked to local fiscal capacity and are widely used to finance infrastructure spending and industrial subsidies.

A distinctive institutional feature is the separation between primary land development and real estate development. Primary development—consisting of land requisition, resettlement, and the provision of basic site infrastructure (Figure 3, Paths 1 and 2)—is typically delegated to LGFVs. Once the land is made ready for construction, land-use rights are transferred to developers through zhao-pai-gua. This chain creates a tight financial nexus: the solvency of LGFVs depends on the market value of the land they prepare, while developers’

demand for land is sensitive to credit availability.

LGFVs function as the off-budget financing arm of local governments. Initially established to circumvent formal prohibitions on municipal borrowing, these corporatized entities raise funds via bank credit and bond issuance (Bai, Hsieh, and Song, 2016; Chen, He, and Liu, 2020; Gao, Ru, and Tang, 2021). Their borrowing capacity is typically supported by land assets injected by local governments and, in many cases, by implicit government guarantees. Importantly, most LGFVs operate under a policy mandate to provide public goods rather than maximizing commercial profit; they generally lack the specialized organizational capabilities, branding, and sales networks required for large-scale housing development.

In contrast to the policy-driven and implicitly supported financing model of LGFVs, real estate developers depend on a far more marketized and financially fragile funding structure. Developers’ financing models are characterized by high leverage and a “high-turnover” strategy that relies heavily on the continuous rollover of debt and customer deposits from pre-sales. By the end of 2020, the sector’s average asset-liability ratio stood at 80.7%, with total liabilities reaching RMB 85.7 trillion. This high degree of financial fragility makes the sector acutely vulnerable to liquidity shocks and regulatory tightening in developer financing.

2.2 “Three-Red-Lines” (TRL) Policy

As discussed in Section 2.1, the real estate sector has accumulated substantial debt, with heavy speculation in the housing market driven by the belief that housing prices would continue to rise. In 2017, President Xi introduced the widely cited slogan “Housing for living, not for speculation”, aiming to tighten the real estate market. To further address these growing risks, Chinese authorities later introduced the TRL policy. This policy was unveiled on August 20, 2020, at a symposium for key real estate enterprises convened by the People’s Bank of China (PBoC) and the Ministry of Housing and Urban-Rural Development, China’s housing regulator. The symposium was attended by 12 major real estate developers, including Country Garden, Evergrande, Vanke, and Sunac.

The TRL policy establishes thresholds for three key financial metrics for property developers: (1) Asset-liability ratio (excluding advance receipts) $\leq 70\%$; (2) Net debt ratio $\leq 100\%$; (3) Cash to short-term debt ratio ≥ 1.0 . Firms that cross one, two, and three of these

thresholds are categorized into “yellow”, “orange”, and “red” tier, respectively, with tiered restrictions on the growth of their interest-bearing liabilities. Companies fully compliant with these thresholds are permitted to increase their debt by a maximum of 15 percent annually. However, any violation of the three red lines reduces the allowed debt growth, which declines by 5 percentage points for each violation, down to zero for firms in the “red tier”.

This policy marks a significant shift in China’s regulatory approach to real estate companies’ debt management. For the first time, clear and specific standards for property developers’ borrowing were established, effectively closing loopholes that companies previously exploited. The policy’s comprehensive scope extends beyond credit bonds to include trust products, asset management products, and overseas financing. Complementing the “Three-Red-Lines,” the “Two-Red-Lines” policy for bank lending was introduced in late 2020, setting upper limits on the proportion of real estate loans and personal housing loans in banks’ portfolios.⁸

Aligning with President Xi’s vision of housing as a basic necessity rather than a speculative asset, the slogan and the TRL policy are well-intentioned to promote housing affordability and maintain financial stability in the real estate sector (Xiong, 2023). They underscore the central government’s commitment to incentivize developers to improve their financial health, reduce leverage, and maintain adequate liquidity, with the broader aim of deleveraging the real estate industry and mitigating financial risks.⁹ However, these policies may have profoundly impacted the Chinese real estate industry. This paper studies how the resulting credit tightening for developers affects outcomes for developers, homebuyers, and

⁸At the end of 2020, the People’s Bank of China (PBoC) and the China Banking and Insurance Regulatory Commission (CBIRC) jointly issued the “Notice on Establishing a Management System for Concentration of Real Estate Loans in Banking Financial Institutions.” In the Notice, the PBoC and the CBRIC set caps on the proportion of real estate loan balance to the balance of all RMB loans (i.e., the real estate loan ratio) and the proportion of personal housing loan balance to the balance of all RMB loans (i.e., the personal housing loan ratio). This regulation categorizes banks into five tiers with different caps: large state-owned banks are limited to 40% for real estate loans and 32.5% for personal housing loans; medium-sized banks to 27.5% and 20% respectively; small banks and non-county rural financial institutions to 22.5% and 17.5%; county-level rural financial institutions to 17.5% and 12.5%; and village banks to 12.5% and 7.5%. See the detailed Notice [here](#).

⁹Another motivation behind this policy is Beijing’s desire to redirect capital from less productive sectors, such as real estate, towards more productive industries, particularly high-end manufacturing. This shift aims to upgrade China’s value chain. Figure IA1 in the Internet Appendix demonstrates a significant increase in loans to industrial sector following the implementation of the TRL policy. See also the discussion among investment professionals [here](#).

local governments’ land finance.

3 Data and Summary Statistics

In this section, we describe the four datasets used in this paper and present their summary statistics: (1) transaction-level data on land acquisitions in the primary land market; (2) financial information for publicly listed real estate firms; (3) project-quarter-level housing sales data; and (4) a list of Local Government Financing Vehicles (LGFVs) and their subsidiaries.

3.1 Land Transaction Data

We obtain transaction-level data on commercial and private residential land parcels in the primary land market from the [CRIC information system](#). We rely primarily on CRIC because developers often acquire land through project subsidiaries; CRIC systematically maps such subsidiary-level acquisitions to their ultimate parent firms, allowing us to accurately link land parcels to publicly listed real estate developers. For each land transaction, we have detailed characteristics, including land parcel size (in hectares), detailed address, plot ratio, land quality, and land use type, alongside the transaction value and buyer’s identity. Moreover, CRIC also records the buyer’s (i.e., developer) equity share in each land acquisition. We cross-checked the coverage of the CRIC data with the data from [Land China](#), maintained by the Ministry of Natural Resources ([Chen and Kung, 2019](#)). We find that, from 2013 to 2023, CRIC covers around 90% of land parcels, 90% of land area, and nearly 100% of the land value covered by Land China. See Table [IA1](#) in the Internet Appendix for a detailed comparison.

Figure [2](#) displays the trends in total land area and total land value of private housing and business lands in the primary land market from 2017 to 2023. The figure shows that land transactions peaked in 2020 and experienced a significant decline in 2022 and 2023.

[Insert Figure [2](#) about here]

3.2 Public Real Estate Firms

Our sample comprises real estate companies listed on the A-share and Hong Kong Stock Exchange, with data sourced from WIND. The initial sample included 108 A-share and 272 Hong Kong-listed firms. We excluded four dual-listed companies to avoid double counting. To ensure that firms are actively engaged in property development, we further filtered the sample by matching firms with land acquisition records in the CRIC database. This step is essential because some firms classified as “real estate”—such as “515J Holding” (000560.SZ) and “Emperor International Holdings” (0163.HK)—primarily focus on property management, brokerage, or operations limited to the Hong Kong market, while others function as REITs. By requiring a match with CRIC records, we successfully isolated 160 firms (80 A-share and 80 Hong Kong-listed) that participate in real estate development in mainland China. Besides the financial statement data, we collected the initial default date for both domestic and overseas bonds of real estate enterprises from [Inno Dealing](#).

Panel A of Table 1 shows the summary statistics of the firm-year data for 160 public real estate firms from 2017 to 2023. The sample firms have an average liability-to-asset ratio of 0.66 and an average debt-to-equity ratio of 1.01. The cash-to-short term debt ratio has a mean of 4.62 and a median of 1.13, suggesting that the typical firm holds relatively modest cash reserves compared with short-term debt. *LinesCrossed* is a count variable equal to the number of red lines crossed by a firm at the end of 2019. On average, firms violated 1.23 red lines. *Post* is a dummy variable that equals one for years 2021 and later, and zero otherwise.

[Insert Table 1 about here]

3.3 Project Level Housing Sales Data

We obtain the housing project sales data from the [China Real Estate Index System \(CREIS\)](#) database. For each housing project, CREIS provides the names of the real estate firms that developed the project and the project’s quarterly sales information, including the number of sales units, sales area, sales value, and the average selling price per square meter.¹⁰ Since

¹⁰Sales data in financial reports may be affected by changes in accounting standards, particularly the new Chinese Accounting Standards for Business Enterprises 14 (CAS 14), issued by the Ministry of Finance.

many real estate projects may be jointly developed by two or more real estate developers, the CREIS database also records each developer’s equity share in each project. Based on this, we calculate the equity-adjusted sales area and sales value for each developer in each project. We merge our public real estate firm list with the CREIS housing sales data based on firm name, and we find sales information for 145 public real estate firms from 2017 to 2023. Panel B of Table 1 shows the summary statistics of the project-quarter data for 15,614 housing projects from 2017 to 2023. On average, each housing project sells 86 units per quarter, with an average selling price of RMB 19,334.

In addition to quarterly sales, CREIS provides project-level opening (for-sale) information, which includes each project’s total number of units, total construction area, and opening date. We use these opening records to construct firm-quarter measures of new project openings, including (i) the number of projects (*# of Projects*) opened for sale by each firm in each quarter, (ii) the total number of units (*# of Units*) opened for sale by firm-quarter, and (iii) the total construction area (*Construction area*) opened for sale by firm-quarter. Panel C of Table 1 presents summary statistics for these firm-quarter opening variables.

3.4 Local Government Financing Vehicles and Their Subsidiaries

We obtain the list of Local Government Financing Vehicles (LGFVs) from WIND, including 3,419 unique firms. We then proceed to trace the equity investments of these LGFVs using the Industrial and Commercial Enterprise Information System, which enables us to identify their subsidiaries: those wholly owned by LGFVs and those formed through joint ventures between LGFVs and other firms with LGFVs owning over 50% of the equity. Subsequently, we merge our LGFVs and subsidiary list with the transaction-level land data based on the purchaser’s identity (i.e., firm name). Out of all land parcels for commercial and residential use, LGFVs directly acquired 4.8% of the land purchases. Furthermore, LGFVs’ subsidiaries (Sub-LGFVs) account for an additional 5.4% of the land purchases. Panel D of Table 1 shows the summary statistics of our LGFV-year data. Following [Geng and Pan \(2024\)](#),

These standards were adopted by overseas listed companies starting in 2018 and by domestic listed companies in 2020, potentially impacting reported sales revenue during our sample period. However, project-level housing sales data from online property contract filings in CREIS remain unaffected by these changes, allowing us to avoid their impact on revenue recognition in our empirical analysis.

bond yield spreads are calculated as the difference between a bond’s observed yield and the corresponding China Development Bank (CDB) yield on the same trading date. For each bond, we use its remaining maturity to identify the two nearest benchmark yields on the CDB yield curve and apply linear interpolation to obtain a maturity-matched benchmark yield. The spread is then defined as the bond’s yield minus this interpolated CDB yield. At the LGFV-year level, the average yield spread is 1.545.

4 Empirical Analyses and Results

4.1 The Impact of TRL Policy on Real Estate Firms

To examine the impact of the TRL policy on real estate firms, we first classify the 160 public real estate firms into two groups based on their financial positions in 2019: firms that did not breach any red lines (“green”) and firms that violated at least one (“red”).¹¹ Panel A of Figure 4 illustrates the trend in interest-bearing debt over time for these two groups of firms. Before 2020, both groups increased their interest-bearing debt steadily, with the line-crossing group growing faster, indicating more aggressive borrowing behavior among noncompliant firms. After 2020, non-violating firms continued expanding debt at a slower pace, while line-crossing firms experienced a visible decline in interest-bearing debt, consistent with the policy’s debt constraints.

Panel B shows the average return on equity (ROE) for the two groups of firms over time. ROE declines substantially among firms that violated the red lines. Panel C depicts the average probability of default over the years. After 2020, real estate developers that violated the red lines exhibited a markedly higher bond default probability than those that did not. Panel D shows land acquisition behavior, revealing that firms with policy violations drastically reduced their land purchases after 2021, indicating the policy’s impact on limiting these firms’ ability to acquire new land. By contrast, firms that met all requirements maintained a relatively stable pace of land acquisition after 2021.

¹¹In Table IA2 of the Internet Appendix, we compare the liability-to-asset ratio, debt-to-equity ratio, and cash-to-short-term debt ratio among firms that violate different numbers of “red” lines and across ownership types.

[Insert Figure 4 about here]

After exploring the unconditional patterns, we move to a formal analysis using a difference-in-differences framework with ordered discrete treatments. We measure treatment intensity by the number of red lines crossed by each real estate developer based on its 2019 financial statements, which provide a clean pre-policy baseline. The TRL policy was first officially communicated to key developers on August 20, 2020; consequently, financial data from 2020 onward may already incorporate anticipatory responses—such as deleveraging, liquidity adjustments, and other balance-sheet management—undertaken in expectation of the forthcoming regulation. Defining treatment intensity using financials after 2020 would therefore contaminate the pre-treatment classification and introduce post-announcement bias into our estimates. The regression specifications are as follows:

$$Y_{it} = \beta(LinesCrossed_i \times Post_t) + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{it} \quad (1)$$

$$Y_{it} = \sum_{\tau=2018}^{2023} \beta_{\tau}(LinesCrossed_i \times \mathbf{1}\{t = \tau\}) + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{it} \quad (2)$$

where i denotes firm, and t denotes year. *LinesCrossed* is a count variable that equals the number of red lines crossed by a firm as of 2019. *Post* is a dummy variable that equals one for years from 2021 onward, and zero otherwise. $\mathbf{X}_{it}$ denotes a vector of firm-year control variables. All specifications include firm fixed effects, which absorb time-invariant firm characteristics, and year fixed effects, which capture common shocks and other time-varying factors shared across firms. Standard errors are clustered at the firm level. In the dynamic specification, we omit 2017 as the reference year; thus, each β_{τ} is interpreted relative to 2017.

Panel A of Table 2 presents the main DID regression results. In column (1), the coefficient of the interaction term *LinesCrossed* $\times$ *Post* is significantly negative ($\beta = -0.106$, t -statistic = -2.51), indicating a significant reduction in interest-bearing debt for firms violating more red lines compared to firms violating fewer or no red lines. The coefficient indicates that violating one additional red line leads to a 10.6% reduction in firms' interest-bearing debt after the implementation of the TRL policy. Column (2) reports the effects on developers'

land purchases. We find that treated firms’ land acquisition value declines significantly after the TRL policy. In terms of the economic magnitude, relative to the sample mean, real estate developers that crossed one additional line reduced land purchase value by 27% ($-3.055 / 11.439$). We do not log-transform firm land value because it is zero for 385 of the 1,113 firm-year observations (i.e., no land acquisitions). As a robustness check, we re-estimate the model using a log specification and obtain qualitatively similar results. Columns (3)–(5) examine the impacts on firms’ sales, financial performance, and credit default probability. The results show that crossing one additional red line leads to a 6.6% reduction in sales, an 8.3-percentage-point reduction in ROE, and a 10.7-percentage-point increase in credit default probability. Overall, these results align with the descriptive patterns in Figure 4.

Panel B of Table 2 shows the dynamic impacts of the TRL policy on real estate firms over 2017–2023, with 2017 as the baseline year. The coefficients for 2018 and 2019 are all insignificantly different from zero. In 2020, we begin to observe moderate declines in interest-bearing debt and ROE, likely due to the fact that the Chinese government announced the TRL policy in August 2020 to 12 major real estate developers, and firms began to adjust their financial strategies preemptively in anticipation of the upcoming restrictions. However, the magnitudes of these declines (i.e., -0.099 and -0.030 in Columns (1) and (4)) are only about half the size of the declines observed in the subsequent years. Consistent with the parallel-trends assumption, the pre-policy coefficients (2018–2020) are jointly insignificant for the other outcomes, and we cannot reject the null of no pre-trends based on the corresponding F -tests (p -values are large).

More specifically, column (1) shows that, relative to the base year of 2017, firms crossing one more red line reduced their interest-bearing debt by 9.9% in 2020, 14.0% in 2021, 14.4% in 2022, and 13.2% in 2023. The results in column (2) show that the post-treatment negative effects of the TRL policy on treated firms’ land acquisition start right after the implementation year 2021, with the decline in land acquisition value becoming statistically significant in 2022 and 2023. Results in column (3) show that crossing one additional red line reduces firms’ sales by 6.0% in 2021, by 6.7% in 2022, and by 5.2% in 2023, although the effects are statistically insignificant. Column (4) shows that line-crossing firms’ ROE also decreases significantly, with one additional red-line violation reducing ROE by 3.0 percentage points

in 2020, 9.0 percentage points in 2021, 11.8 percentage points in 2022, and 8.6 percentage points in 2023. We find in column (5) that line-crossing firms’ credit default probability increased by 5.1 percentage points, 13.0 percentage points, and 14.7 percentage points in the three years following policy implementation. Overall, the TRL policy has an immediate and strong negative impact on line-crossing developers. Both financial performance and land acquisitions of these line-crossing firms are significantly affected by the policy.

[Insert Table 2 about here]

Next, we examine the heterogeneous impact of the TRL policy across different types of firms. In our real estate firm sample, 61% of firms are privately owned and 39% are state owned. In Figure 5, we plot the trends in debt financing, financial performance, probability of default, and land acquisition behavior for state-owned line-crossing developers (dashed red lines) and privately-owned line-crossing developers (solid red lines). We define firm types based on their ownership in 2019. The figure reveals a striking divergence in trajectories between privately-owned and state-owned line-crossing developers following the implementation of the TRL policy. Privately-owned line-crossing developers experienced a more pronounced decline across multiple financial indicators, including a sharp drop in interest-bearing debt, deteriorating financial performance, a dramatic spike in default probability, and a steeper decline in acquired land values. By contrast, state-owned line-crossing developers maintained relatively stable financial positions, with steady debt levels, relatively better performance, and negligible default risk. This differential impact suggests that the TRL policy may have disproportionately affected privately-owned developers, potentially constraining their access to credit, eroding their financial stability, and increasing their vulnerability to market pressures.

[Insert Figure 5 about here]

In Figure 6, we plot trends in debt financing, financial performance, probability of default, and land acquisition behavior for state-owned compliant developers (dashed green lines) and privately owned compliant developers (solid green lines). Across Panels A–D, outcomes for both groups evolve smoothly through the introduction of the TRL policy. These patterns

suggest that compliant firms were largely insulated from the policy and that the salient effects of the TRL are concentrated among line-crossing developers.

[Insert Figure 6 about here]

We next conduct a formal regression analysis on the heterogeneous effects of the TRL policy on real estate developers across ownership types. In Table 3, we include the triple interaction between *Private*, *LinesCrossed*, and *Post* to examine the heterogeneous impacts. In column (1), the coefficients on $LinesCrossed \times Post$ and $Private \times Post$ are significantly negative, while the coefficient on the triple interaction term is insignificant. This pattern indicates that, compared to SOEs, private enterprises experience a larger post-policy decline in interest-bearing debt overall. After accounting for this overall differential trend, we find no significant difference between private enterprises and state-owned enterprises that cross the same number of “red” lines. In column (2), we do not find a statistically significant difference in land purchasing between line-crossing firms with different ownership types. In columns (3)–(5), we observe that violating one more red line reduces privately-owned developers’ sales by 12.8%, ROE by 9.9 percentage points, and increases the probability of default by 15.9 percentage points. Taken together, the results in Table 3 suggest a similar direct effect of the TRL policy on debt financing and land acquisition behavior of line-crossing firms, but a divergence in sales and financial performance between state-owned and privately-owned developers after 2021, highlighting the policy’s unintended uneven effects on the real estate sector.¹²

[Insert Table 3 about here]

4.2 The Impact of TRL Policy on Housing Sales

In this section, we leverage granular project-quarter housing sales data to examine the impact of the TRL policy on housing sales. Specifically, we aggregate the project-quarter data to the

¹²In a related study, Fang and Zhang (2025) document a resurgence of SOEs in the real estate sector following the 2012 anti-corruption campaign, albeit through mechanisms different from those we document for the Three-Red-Lines policy in 2021.

firm-quarter level to estimate the overall effect of the TRL policy on housing sales volumes and prices for public real estate firms.

We begin with the aggregate analysis by constructing four key firm-quarter outcome variables. $\ln(\text{Sales units})$ is the natural logarithm of the total number of housing units sold by a firm in a given quarter. Similarly, $\ln(\text{Sales area } (1k \text{ m}^2))$ is the natural logarithm of the total equity-adjusted sales area (in thousands of square meters), while $\ln(\text{Sales value } (1m \text{ RMB}))$ is the natural logarithm of the equity-adjusted sales value (in millions of RMB) in a given quarter. Finally, $\ln(\text{Sales price } (RMB/m^2))$ represents the natural logarithm of the average sales price per square meter of housing units sold by the firm in a quarter.

Panel A of Table 4 presents the overall post-treatment effects. The TRL policy leads to significant declines in housing sales among public real estate firms. Specifically, each additional red line violation is linked to a 12.4% decline in the number of units sold, a 12.8% decrease in sales area, a 17.1% decline in sales value, and a 4.3% drop in average housing prices during the post-TRL period.

[Insert Table 4 about here]

We further examine the heterogeneous effects of the TRL policy on housing sales across different types of real estate firms. Specifically, we include a triple interaction term between *Private*, *LinesCrossed*, and *Post*, to assess whether privately owned developers are more affected by TRL violations. Panel B of Table 4 presents the regression results. Columns (1)–(3) show that line-crossing private developers experience significantly larger reductions in sales units, sales area, and sales value than their state-owned counterparts. In terms of economic magnitude, relative to state-owned developers violating the same number of red lines, privately owned developers sold 16.9% fewer units, 21.0% less housing area, and 21.8% less sales value, indicating a stronger policy impact on private firms. Column (4) reveals that the sales price drop is similar for state-owned and private developers, suggesting that the larger decline in sales value is not driven by steeper price cuts but instead reflects lower sales volume (units and area). Overall, the firm-quarter results reinforce our earlier findings from firm-year financial data: TRL violations lead to significant sales and financial deterioration, with disproportionately stronger effects on privately owned developers.

4.3 The Impact of TRL Policy on the Housing Supply and Demand

To investigate whether the decline in housing sales is driven by a contraction in housing supply or weaker homebuyer demand, we conduct two sets of tests.

First, we measure developers' housing supply using the opening-for-sale of new housing projects. The CREIS database reports each project's first opening-for-sale date, the number of units opened, and total construction area. We aggregate these project-level data to the firm-quarter level and examine how the TRL policy affects the supply of newly opened projects.

Panel A of Table 5 shows the average treatment effects of the TRL policy on the supply of housing by public real estate firms. Across specifications, more exposed firms contract supply in a pronounced and statistically significant manner. In column (1), the coefficient on $LinesCrossed \times Post$ implies that crossing one additional red line is associated with a 25.5% ($\exp(-0.295) - 1$) decline in the number of projects opened for sale in the post-policy period. Columns (2) and (3) show comparable contractions in scale: units and construction area fall by 35.1% ($\exp(-0.432) - 1$) and 33.8% ($\exp(-0.412) - 1$), respectively. Taken together, these results indicate that the TRL policy materially curtailed developers' capacity and willingness to initiate new housing supply.

Panel B of Table 5 examines heterogeneity by ownership type by augmenting the baseline specification with the triple interaction $Private \times LinesCrossed \times Post$. The estimates indicate that privately owned developers contract supply significantly more than state-owned developers following the TRL policy, conditional on the extent of TRL violations. The triple-interaction coefficients are negative and statistically significant across all three measures of new supply, implying larger post-policy declines for private firms in the number of projects, units, and construction area. Overall, the evidence reinforces the view that the TRL policy's tightening effects are borne disproportionately by private developers.

[Insert Table 5 about here]

Having shown that the TRL policy leads more exposed developers to open fewer and smaller new projects, we next ask whether the policy also depresses housing demand for

projects that remain on the market. To isolate demand-side changes from supply-side adjustments operating through the extensive margin, we exploit within-project variation in sales outcomes over time. Specifically, we estimate project-quarter regressions with project fixed effects, so identification comes from changes in sales for the same project before and after the policy, holding constant all time-invariant project characteristics. This design isolates demand on the intensive margin by testing whether sales for the same project decline after the TRL policy, conditional on the project remaining on the market.

Panel A of Table 6 reports project-quarter regressions that exploit within-project variation by including project fixed effects, while absorbing time-varying local demand conditions with city-year-quarter fixed effects. The city-year-quarter fixed effects are particularly important in our setting because, during the sample period, many cities adjusted housing-market policies (e.g., easing purchase restrictions), which could directly affect demand. By comparing projects within the same city and year-quarter, we effectively net out the influence of these concurrent local policy shifts.¹³ Across all four outcomes—log sales units, log sales area, log sales value, and log price—the coefficient on $LinesCrossed \times Post$ is economically small and statistically indistinguishable from zero, suggesting a limited average demand-side response conditional on a given project being on the market.

Panel B of Table 6 explores heterogeneity by ownership type by adding the triple interaction $Private \times LinesCrossed \times Post$. The estimates reveal a sharp divergence across developer types. The triple interaction estimates are negative and statistically significant for log sales units, log sales area, and log sales value, indicating that greater TRL exposure leads to larger post-policy declines in within-project sales for private developers relative to SOEs. At the same time, the coefficient on $LinesCrossed \times Post$ is positive and significant for the baseline group (state-owned developers), implying that SOE projects experience higher sales in the post-policy period as exposure increases. Taken together, the opposing signs are consistent with a reallocation of demand away from private developers’ projects and toward SOE-developed projects after the TRL policy.

[Insert Table 6 about here]

¹³We obtain similar results if we control for the more stringent block-year-quarter fixed effects. A block corresponds to a within-city real estate submarket (e.g., a neighborhood-level housing market segment commonly used in the property market). The results are reported in Table IA3 of the Internet Appendix.

In sum, the TRL policy’s direct effect operates primarily through the supply side. More exposed developers open fewer new projects and bring fewer units and less construction area to market. At the same time, the policy generates meaningful heterogeneity across firms, with both supply- and demand-side forces at work. On the supply side, private developers reduce new-project supply more sharply; on the demand side, sales within continuing projects shift away from private developers and toward SOEs, consistent with homebuyers reallocating demand from POEs to SOEs in China’s housing presale system, where concerns about project completion risk are more salient. Taken together, the TRL policy appears to have constrained developers’ expansion while simultaneously reshaping competitive dynamics in favor of state-owned firms.

4.4 The Spillover Effects on Land Market and Local Government Financing

Real estate firms tend to maintain operations in cities where they have an existing presence. Our data show that, on average, 63% of the land acquired by real estate firms is located in cities where the same firms have purchased land within the past three years. This purchasing inertia motivates the construction of a city-level intensity measure to capture each city’s exposure to the TRL policy using land transaction data from the pre-policy period (2017–2019). Specifically, the city-level TRL policy intensity measure (*Intensity*) is defined as the ratio of the land transaction value purchased by red-line-violating firms to the total land transaction value purchased by all listed firms in a city over 2017–2019. A higher *Intensity* reflects greater local exposure to the TRL policy.

Panel A of Figure 7 provides descriptive evidence on how land markets shifted after the introduction of the TRL policy. Before 2020, real estate firms consistently dominated the land market, purchasing far more land than LGFVs and Sub-LGFBs combined. This pattern weakened sharply after the policy shock. Real estate firms’ total land purchases fell from RMB 2.63 trillion in 2020 to RMB 0.74 trillion in 2023. By contrast, purchases by government-linked entities expanded, especially among Sub-LGFBs, whose land acquisitions increased from RMB 0.32 trillion in 2020 to RMB 0.69 trillion in 2022. By 2022, LGFBs

and Sub-LGFVs together were purchasing more land than listed real estate firms.

This shift may have important fiscal consequences if land acquired by LGFVs and Sub-LGFVs is less likely to be developed than land acquired by listed developers. To examine this issue, we use housing project-level opening-for-sale data from CREIS and identify 35,529 projects with sufficient information for land-to-project matching. We link each land parcel to a project based on developer identity, location, and a plausible timeline in which the project launch follows the land acquisition. Parcels without a corresponding project match are classified as undeveloped, whereas matched parcels are classified as developed.

We then aggregate these parcel-level measures to the land-acquisition-year level and report, for each year, the total acquisition value of parcels classified as undeveloped. Panel B of Figure 7 reports these values. The key result is that the post-2021 increase in land purchases by LGFVs and Sub-LGFVs translated into a large and rising stock of undeveloped land. For Sub-LGFVs, the annual average undeveloped land value increased by 148%, from RMB 176.4 billion in 2017–2020 to RMB 437.6 billion in 2021–2023. Figure 8 further shows that development rates declined markedly. The LGFV development ratio fell from an average of 14.6% in 2017–2019 to 5.2% in 2021 and 2.1% in 2023, while the Sub-LGFV development ratio declined from roughly 12.0% in 2017–2019 to 8.3% in 2021 and 4.6% in 2023.

[Insert Figures 7 and 8 about here]

Table 7 reports the results for the effects of the TRL policy on LGFV propping activities and financial conditions. Panel A presents the effects on LGFV propping activities. The dependent variable, *Propping ratio (area)*, is defined as the proportion of residential land area purchased by LGFVs or their controlled subsidiaries that remains undeveloped through 2025. The ratio is set to zero when no such purchase occurs, and is also coded as zero in the pre-policy period. The *Propping ratio (value)* is defined similarly using the land value. The positive and statistically significant coefficient on $Intensity \times Post$ indicates that LGFVs in cities more exposed to the TRL policy increased their land-market propping activity relative to those in less exposed cities. The magnitude is also economically meaningful: given that the standard deviation of *Intensity* is 0.156, a one-standard-deviation increase in policy exposure implies an increase of about 0.047 in the propping ratio, which is equivalent to

17.4% of the sample mean of the propping ratio.

Panel B presents the effects on LGFV financial conditions. Financing costs are captured by the *LGFV yield spread*, measured as the average yield spread at close for all outstanding bonds issued by the LGFV. We use three variables (*Current ratio*, *Quick ratio*, and *Cash ratio*) to capture firms’ short-term liquidity. The coefficient on $Intensity \times Post$ is positive and statistically significant in the yield spread regression, and negative and statistically significant in the liquidity ratio regressions. These results indicate that LGFVs located in cities with greater exposure to the TRL policy experienced a deterioration in financial conditions in the post-policy period, as reflected in higher financing costs and weaker short-term liquidity. The magnitudes are also economically meaningful. Given that the standard deviation of *Intensity* is 0.156, a one-standard-deviation increase in policy exposure implies an increase of about 0.230 in LGFV yield spread, which is equivalent to 14.9% of the sample mean. The corresponding declines are about 0.290 for the current ratio, 0.145 for the quick ratio, and 0.051 for the cash ratio, equivalent to 7.7%, 9.5%, and 13.1% of their respective sample means. Table [IA4](#) further shows that greater LGFV propping activity is associated with weaker financial conditions.

[Insert Table [7](#) about here]

Taken together, our evidence suggests that the TRL policy led to a sharp contraction in land acquisitions by real estate firms, while LGFVs—especially Sub-LGFVs—partly stepped in to absorb land supply and stabilize the land market. However, a substantial share of the land acquired by these government-linked entities was not subsequently brought to market, which may increase local governments’ off-balance-sheet debt risks. These patterns point to a meaningful post-policy reallocation of land demand toward LGFVs and their subsidiaries, shedding light on China’s evolving fiscal governance.

5 Conclusion

This paper examines the economic consequences of China’s TRL policy, focusing on its impact on real estate firms, local government financing, and the land market. Our analysis

reveals that the policy’s strict financial constraints significantly curtailed land purchases and financial leverage among real estate firms, a likely first-order cause of China’s recent economic downturn. This contraction was partly offset by a strategic shift in land purchases by LGFVs and, more prominently, Sub-LGFVs, which have become pivotal buyers in the land market. However, land acquired by (Sub-)LGFVs after 2021 was developed much more slowly than land acquired by non-LGFV entities, which may have temporarily propped up reported land-sale revenues while increasing local governments’ off-balance-sheet debt risks.

Our study fills a critical gap in the literature by documenting the far-reaching consequences of the TRL policy, including the transformation of real estate firms’ behavior and the growing importance of (Sub-)LGFVs in the land market. This nuanced understanding provides valuable insights for policymakers, market participants, and scholars, offering a comprehensive perspective on the interconnected dynamics of real estate, land markets, and local government financing in China’s evolving economic landscape.

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Appendix Table A1: Variable Definitions

Table A1: Variable Definitions

Variable name	Definition
LinesCrossed	Count variable that equals the number of red lines a firm crossed at the end of 2019
Post	Indicator that equals one for years on and after 2021
Private	Indicator that equals one if a firm is a privately-owned enterprise
Size	Firm size is measured as the natural logarithm of total assets
Liability-to-asset ratio	The ratio of total liability excluding advance payments from customers to total assets excluding advance payments
Debt-to-equity ratio	Interest-bearing debt minus cash and cash equivalents, divided by total equity
Cash-to-short term debt ratio	The sum of cash and cash equivalents, divided by current liabilities
Ln(Interest-bearing debt)	The natural logarithm of the interest-bearing debt
Ln(Sales)	The natural logarithm of sales revenue
ROE	Net profit scaled by the average total equity in this year and last year
Default	Indicator that equals one if a firm has experienced a credit default in a year or previous years
Firm land value (1b RMB)	The total land value (in billions RMB) of all lands purchased by the firm
Sales units	The number of housing units sold
Sales area (1k m^2)	The equity-adjusted sales area in thousands of square meters sold
Sales value (1m RMB)	The equity-adjusted sales value in millions of RMB sold
Sales price (RMB/ m^2)	The average selling price per square meter of housing units sold
# of Projects	The number of new housing projects opened for sale
# of Units	The equity-adjusted total number of units of new housing projects opened for sale
Construction area (10k m^2)	The equity-adjusted total construction area of new housing projects opened for sale
Propping ratio (area or value)	Propping ratio is defined as the proportion of residential land (area or value) purchased by LGFVs or their controlled subsidiaries that remains undeveloped through 2025. The ratio is set to zero when no such purchase occurs, and is also coded as zero in the pre-policy period
LGFV yield spread	The average yield spread at close for all outstanding bonds issued by the LGFV
Current ratio	Current assets divided by current liabilities
Quick ratio	Current assets minus inventories, divided by current liabilities
Cash ratio	Cash and cash equivalents plus financial assets held for trading, divided by current liabilities
Intensity	The ratio of total value of lands purchased by line-crossing public real estate firms to the total value of lands purchased by all public real estate firms in a given city over the period 2017–2019

This table provides the definitions of the main variables used in our analysis.

Figures

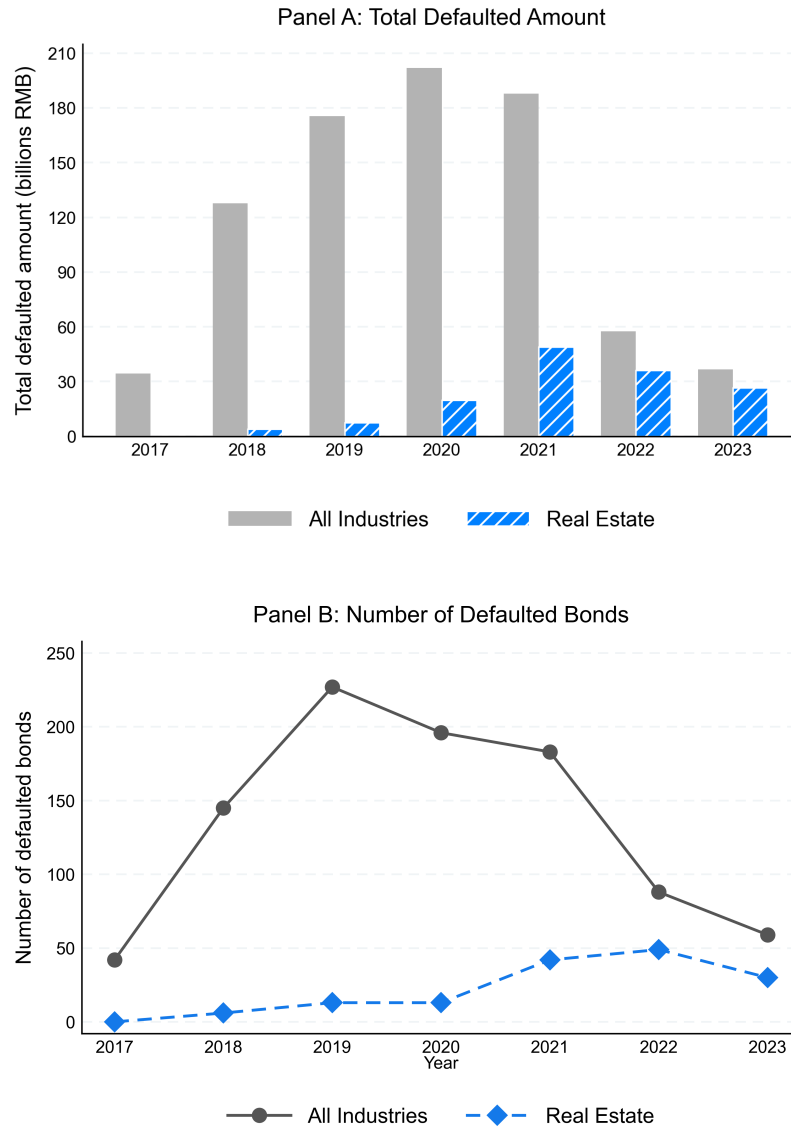


Figure 1: Annual Defaults in China's Corporate Bond Market

This figure presents default activity in China's corporate bond market from 2017 to 2023, separately for the overall market and the real estate sector. Panel A reports the total defaulted amount, measured as the outstanding bond balance on the default date (billions of RMB). Panel B reports the number of defaulted bonds. We exclude default events classified as principal or interest extensions, technical defaults, and cross defaults. We also exclude duplicate bonds with the same bond code across trading venues to avoid double-counting bonds traded in both the exchange and interbank markets.

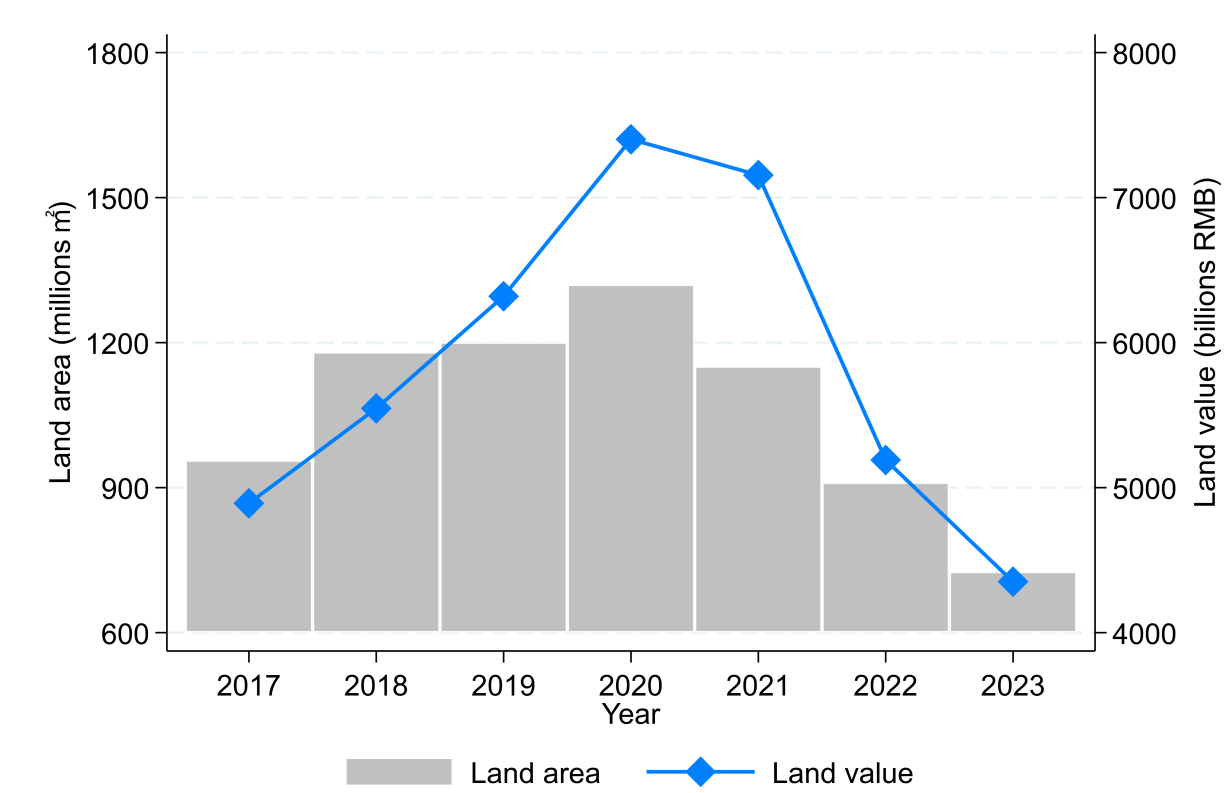


Figure 2: Private Residential and Commercial Land Transactions

This figure reports annual transactions of private residential and commercial land in China's primary land market from 2017 to 2023. Bars represent the total transacted land area (millions of m^2 on the left axis), and the line represents the total transaction value (billions of RMB on the right axis).

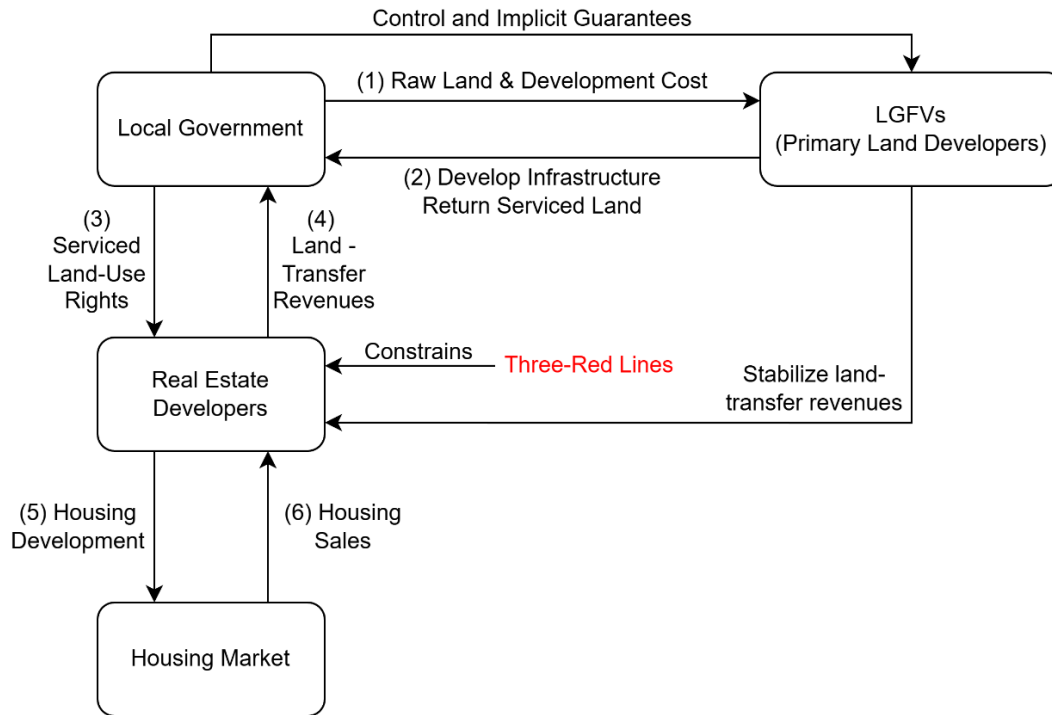


Figure 3: Land-Housing Market Structure: Local Governments, LGFVs, and Developers

This figure summarizes the institutional relationships among local governments, LGFVs (primary land developers), and real estate developers, and illustrates how the Three-Red-Lines policy constrains real estate developers and affects local governments' land-transfer revenues.

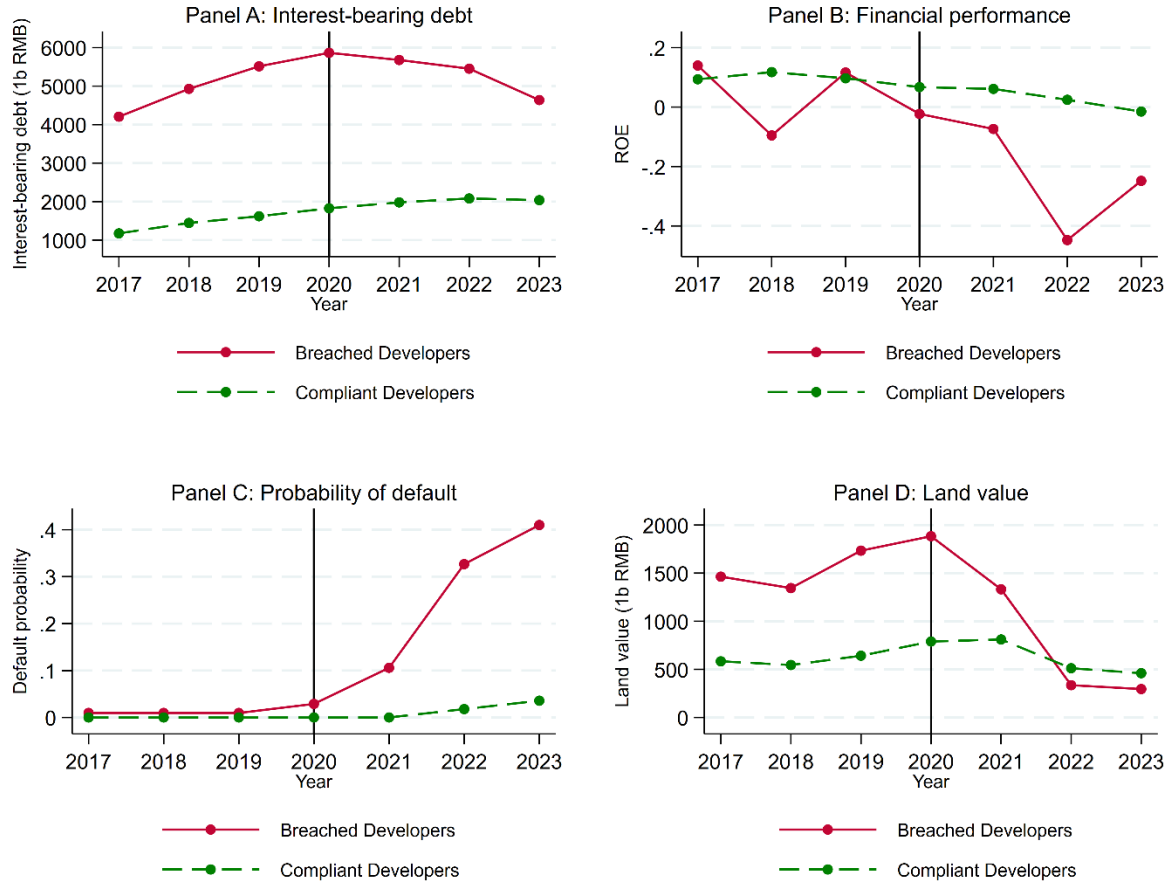


Figure 4: The Impact of Three-Red-Lines on Real Estate Developers

This figure shows the effects of the TRL policy on real estate developers. Developers are classified using their 2019 financial information. Breached developers (solid red line) violate at least one TRL threshold in 2019; compliant developers (dashed green line) violate none. Panels A–D show aggregate interest-bearing debt (billions of RMB), average ROE, average default probability, and aggregate land acquisition value (billions of RMB) for these two groups of developers, respectively.

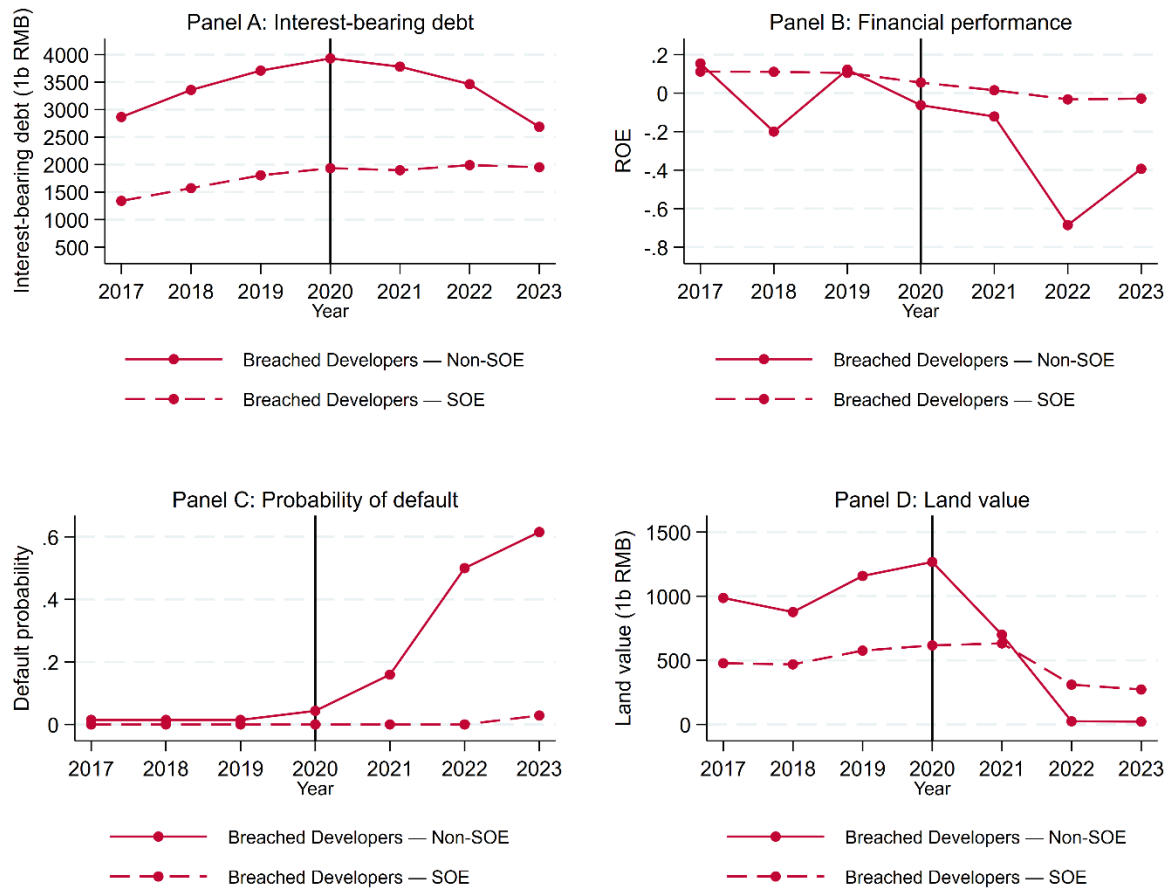


Figure 5: SOEs versus Non-SOEs among Breached Developers

This figure compares SOEs and non-SOEs among breached developers, defined as developers that breach at least one TRL threshold in 2019. The solid red line represents privately-owned listed developers, while the dashed red line represents state-owned listed developers. Panels A–D show aggregate interest-bearing debt (billions of RMB), average ROE, average default probability, and aggregate land acquisition value (billions of RMB) for these two groups of firms, respectively.

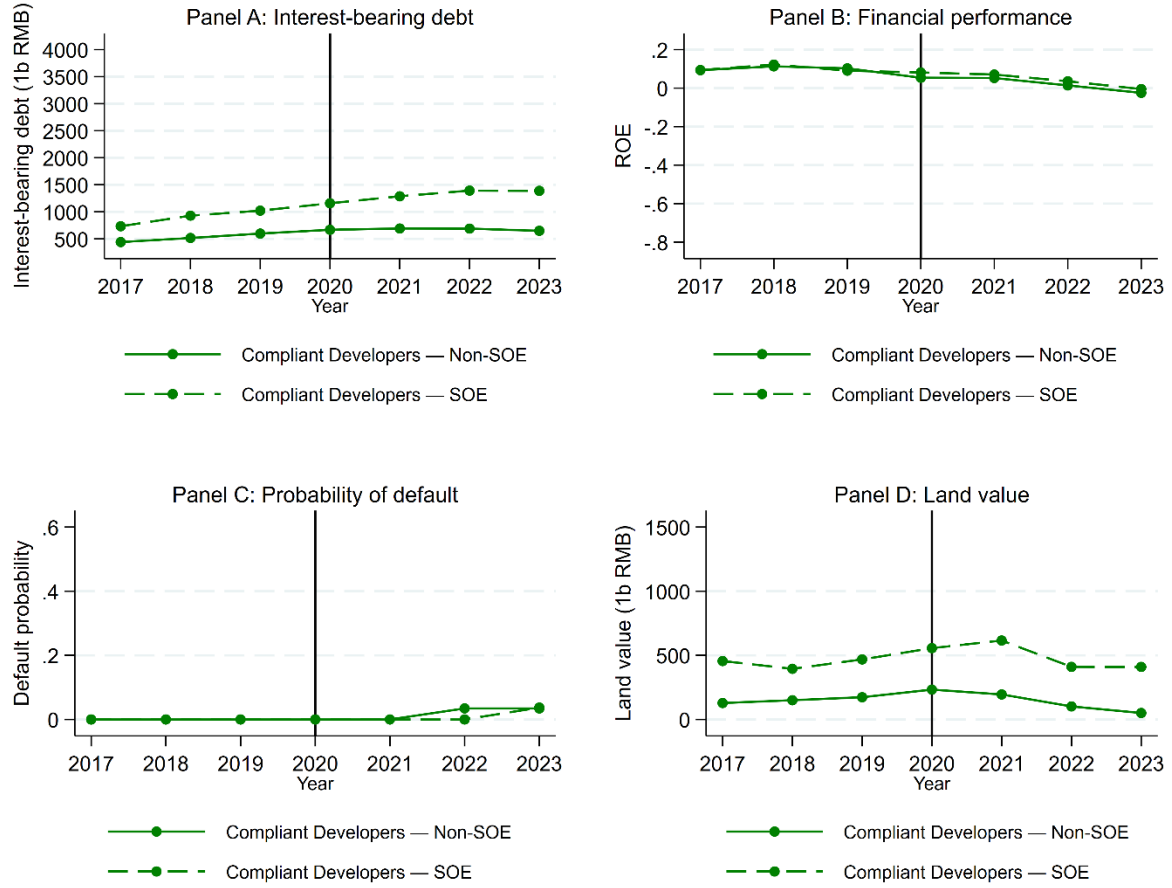


Figure 6: SOEs versus Non-SOEs among Compliant Developers

This figure compares SOEs and non-SOEs among compliant developers, defined as developers that do not violate any TRL threshold in 2019. The solid green line represents privately-owned listed developers, while the dashed green line represents state-owned listed developers. Panels A–D show aggregate interest-bearing debt (billions of RMB), average ROE, average default probability, and aggregate land acquisition value (billions of RMB) for these two groups of firms, respectively.

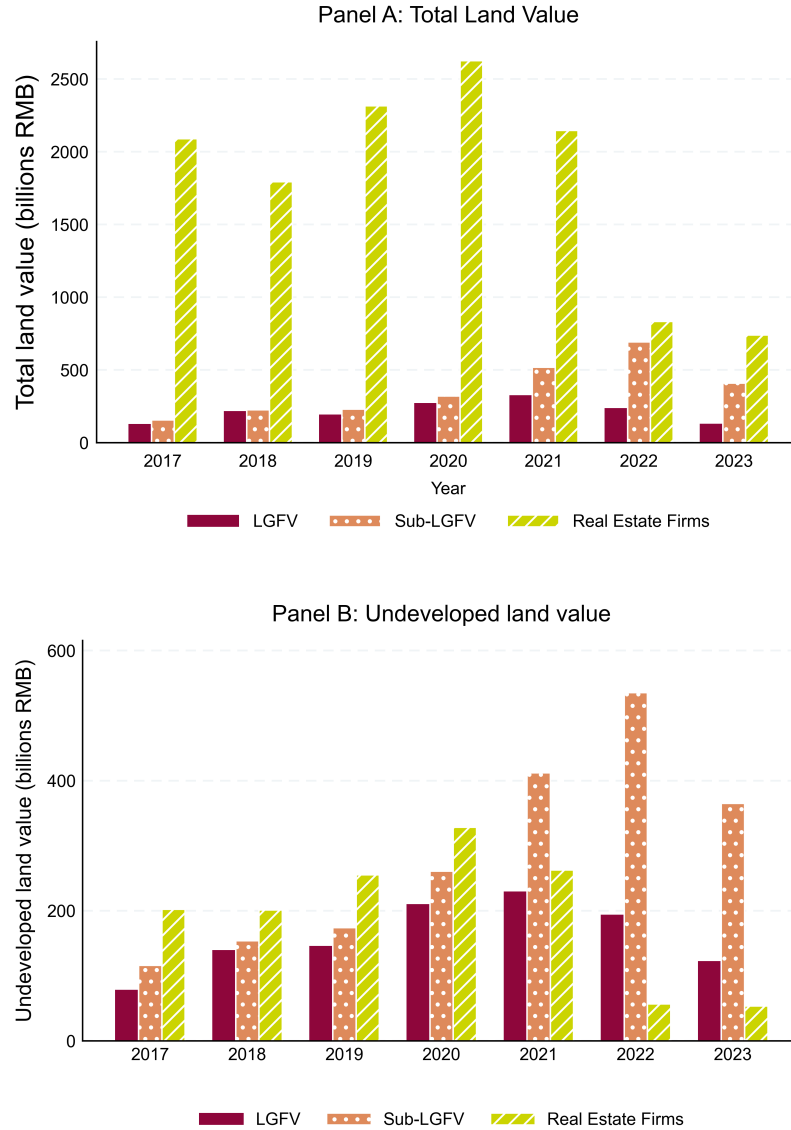


Figure 7: Land Purchases of LGFVs, Sub-LGfVs, and Real Estate Firms

This figure presents the annual value of land acquisitions in China's primary land market for three groups of purchasers: LGFVs (local government financing vehicles), Sub-LGfVs (majority-owned subsidiaries of LGFVs), and publicly listed real estate firms. Panel A shows total land value, and Panel B shows the value of undeveloped land. A parcel is classified as developed if it is subsequently linked to a housing project that opened for sale; otherwise, it is classified as undeveloped.

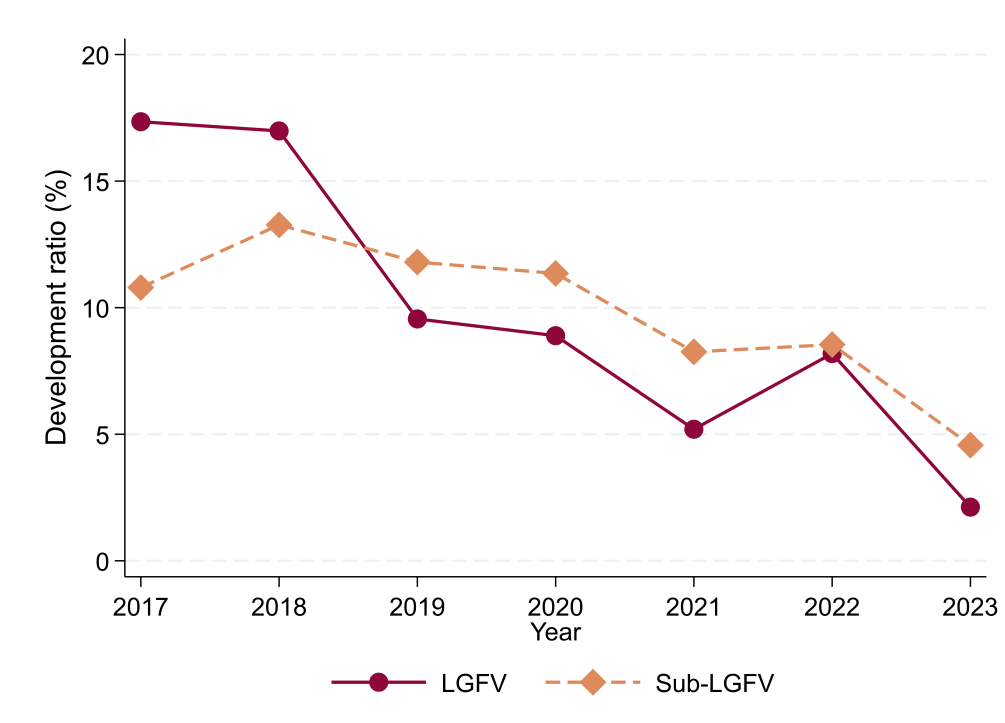


Figure 8: Undeveloped Land Purchases of LGFVs, Sub-LGFVs, and Real Estate Firms

This figure reports the annual development ratio for land acquired by LGFVs and Sub-LGFVs in China's primary land market. The development ratio is measured as the fraction of purchased land parcels that are subsequently linked to a housing project opened for sale.

Tables

Table 1: Summary Statistics

Panel A: Firm-year level financial information

Variables	N	Mean	Std Dev	P5	Median	P95
LinesCrossed	1,113	1.228	1.156	0.000	1.000	3.000
Post	1,113	0.425	0.495	0.000	0.000	1.000
Private	1,113	0.610	0.488	0.000	1.000	1.000
Size	1,113	24.753	1.611	21.869	24.835	27.217
Liability-to-asset ratio	1,103	0.660	0.199	0.260	0.714	0.908
Debt-to-equity ratio	1,103	1.008	1.458	-0.344	0.691	3.485
Cash-to-short term debt ratio	1,076	4.620	16.793	0.060	1.133	12.152
Ln(Interest-bearing debt)	1,087	23.445	1.977	20.074	23.753	25.979
Firm land value (1b RMB)	1,113	11.439	24.23	0.000	1.397	61.243
Ln(Sales)	1,103	22.991	1.715	20.054	23.056	25.725
ROE	1,092	0.029	0.203	-0.406	0.063	0.243
Default	1,113	0.084	0.278	0.000	0.000	1.000

Panel B: Project-quarter level sales data

Variables	N	Mean	Std Dev	P5	Median	P95
Sales units	181,708	86.058	145.915	1.000	34.000	340.000
Sales area (1k m^2)	181,708	7.345	13.743	0.087	2.602	29.947
Sales value (1m RMB)	181,708	121.532	260.164	1.112	40.278	492.790
Sales price (RMB/ m^2)	181,689	19334.160	16023.89	6255.712	14475.888	47912.805

Panel C: Firm-quarter level new project opening data

Variables	N	Mean	Std Dev	P5	Median	P95
# of Projects	3,892	4.123	10.388	0.000	1.000	20.000
# of Units	3,892	4806.096	22976.539	0.000	183.510	20417.550
Construction area (10k m^2)	3,892	107.968	787.914	0.000	3.415	417.563

Panel D: LGFV-year level data

Variables	N	Mean	Std Dev	P5	Median	P95
Propping ratio (area)	7,139	0.271	0.443	0.000	0.000	1.000
Propping ratio (value)	7,139	0.269	0.442	0.000	0.000	1.000
LGFV yield spread	7,139	1.545	1.531	0.000	1.112	4.384
Current ratio	7,139	3.766	2.772	1.136	3.011	9.024
Quick ratio	7,139	1.522	1.134	0.402	1.234	3.692
Cash ratio	7,139	0.385	0.372	0.048	0.280	1.101
Intensity	7,139	0.839	0.156	0.586	0.863	1.000

This table reports the summary statistics for different sets of data used in our analysis. Panel A reports the firm-year level data consisting of 160 public real estate firms from 2017 to 2023. Panel B presents the project-quarter level housing sales data from the China Real Estate Index System (CREIS) database, consisting of 15,614 housing projects of 145 public real estate firms from 2017 to 2023. Panel C presents the firm-quarter level new housing project opening data of public real estate firms from 2017 to 2023. Panel D reports summary statistics for LGFV-year level variables, including measures of LGFVs' propping behavior, bond yield spreads and financial ratios, over the 2017 to 2023 period. Variable definitions are provided in Appendix Table A1. All continuous variables are winsorized at the 1st and 99th percentiles.

Table 2: The Impact of Three-Red-Lines Policy on Real Estate Firms

Panel A: Overall effects

Variables	(1) Ln(Interest-bearing debt)	(2) Firm land value (1b RMB)	(3) Ln(Sales)	(4) ROE	(5) Default
LinesCrossed $\times$ Post	-0.106** (-2.51)	-3.055*** (-2.81)	-0.066** (-1.99)	-0.083*** (-6.03)	0.107*** (5.14)
Size	0.681*** (5.94)	0.007 (0.01)	0.737*** (8.18)	0.002 (0.09)	0.020 (0.69)
Observations	1,087	1,113	1,103	1,092	1,113
Adjusted R^2	0.935	0.775	0.954	0.467	0.427
Firm FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm	Firm

Panel B: Dynamic effects

Variables	(1) Ln(Interest-bearing debt)	(2) Firm land value (1b RMB)	(3) Ln(Sales)	(4) ROE	(5) Default
LinesCrossed $\times$ Year2018	0.002 (0.05)	-0.794 (-0.83)	-0.003 (-0.11)	-0.017 (-1.50)	-0.002 (-0.86)
LinesCrossed $\times$ Year2019	-0.032 (-0.89)	0.396 (0.31)	0.027 (0.97)	-0.014 (-1.20)	-0.003 (-0.89)
LinesCrossed $\times$ Year2020	-0.099*** (-2.78)	-0.219 (-0.19)	0.001 (0.01)	-0.030** (-2.12)	0.008 (0.80)
LinesCrossed $\times$ Year2021	-0.140*** (-3.42)	-2.581 (-1.63)	-0.060 (-1.54)	-0.090*** (-5.20)	0.051** (2.47)
LinesCrossed $\times$ Year2022	-0.144*** (-2.82)	-3.787** (-2.53)	-0.067 (-1.41)	-0.118*** (-6.00)	0.130*** (4.63)
LinesCrossed $\times$ Year2023	-0.132** (-1.98)	-3.296** (-2.26)	-0.052 (-1.11)	-0.086*** (-4.29)	0.147*** (5.10)
Size	0.694*** (5.92)	-0.042 (-0.03)	0.735*** (7.99)	0.006 (0.37)	0.026 (0.84)
Observations	1,087	1,113	1,103	1,092	1,113
Adjusted R^2	0.935	0.774	0.954	0.470	0.439
Firm FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm	Firm
Pre-trends F-test (p -value)	0.002	0.319	0.619	0.164	0.363

This table reports the OLS regression results on the impact of the TRL policy on real estate firms. The sample period is from 2017 to 2023. Panel A shows the overall treatment effects of the TRL policy. Panel B shows the dynamic effects of the policy. Variable definitions are provided in Appendix Table A1. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Pre-trends F-test (p -value) reports p -values from an F-test that the pre-policy interaction coefficients are jointly equal to zero (baseline year: 2017).

Table 3: Differential Effects of the Three-Red-Lines Policy Across Firm Types

Variables	(1) Ln(Interest- bearing debt)	(2) Firm land value (1b RMB)	(3) Ln(Sales)	(4) ROE	(5) Default
Private $\times$ LinesCrossed $\times$ Post	0.069 (0.86)	-2.413 (-1.16)	-0.128** (-2.14)	-0.099*** (-4.82)	0.159*** (5.59)
LinesCrossed $\times$ Post	-0.136*** (-3.11)	-1.380 (-0.96)	0.015 (0.37)	-0.024** (-2.02)	0.003 (0.41)
Private $\times$ Post	-0.369** (-2.19)	-3.807 (-1.34)	-0.077 (-0.74)	-0.039 (-1.52)	0.070* (1.88)
Size	0.652*** (6.15)	-0.379 (-0.35)	0.726*** (8.39)	-0.004 (-0.22)	0.032 (1.05)
Observations	1,087	1,113	1,103	1,092	1,113
Adjusted R^2	0.936	0.780	0.955	0.526	0.514
Firm FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm	Firm

This table reports the OLS regression results on the heterogeneous impact of the TRL policy on real estate firms. The sample period is from 2017 to 2023. Variable definitions are provided in Appendix Table A1. Standard errors are clustered at the firm level, and t -statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4: The Impact of Three-Red-Lines Policy on Housing Sales

Panel A: Post-treatment effects

Variables	(1) Ln(Sales units)	(2) Ln(Sales area (1k m^2))	(3) Ln(Sales value (1m RMB))	(4) Ln(Sales price (RMB/ m^2))
LinesCrossed $\times$ Post	-0.124** (-2.27)	-0.128** (-2.53)	-0.171*** (-3.20)	-0.043** (-2.36)
Size	0.623*** (3.06)	0.560*** (3.00)	0.639*** (3.13)	0.079** (2.12)
Observations	3,883	3,883	3,883	3,883
Adjusted R^2	0.857	0.859	0.843	0.733
Firm FEs	YES	YES	YES	YES
Year-Quarter FEs	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm

Panel B: Differential effects across firm types

Variables	(1) Ln(Sales units)	(2) Ln(Sales area (1k m^2))	(3) Ln(Sales value (1m RMB))	(4) Ln(Sales price (RMB/ m^2))
Private $\times$ LinesCrossed $\times$ Post	-0.169* (-1.71)	-0.210** (-2.34)	-0.218** (-2.37)	-0.008 (-0.20)
LinesCrossed $\times$ Post	0.007 (0.09)	0.023 (0.34)	-0.008 (-0.11)	-0.031 (-0.94)
Private $\times$ Post	-0.367* (-1.80)	-0.233 (-1.27)	-0.364* (-1.90)	-0.131* (-1.78)
Size	0.604*** (3.19)	0.550*** (3.14)	0.620*** (3.30)	0.071** (2.13)
Observations	3,883	3,883	3,883	3,883
Adjusted R^2	0.862	0.863	0.849	0.737
Firm FEs	YES	YES	YES	YES
Year-Quarter FEs	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm

This table reports the OLS regression results on the impact of the TRL policy on housing sales of public real estate firms covered by the CREIS Database. The data is at the firm-year-quarter level. The sample period is from 2017 to 2023. Variable definitions are provided in Appendix Table A1. Standard errors are clustered at the firm level, and t -statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5: The Impact of Three-Red-Lines Policy on Housing Supply

Panel A: Post-treatment effects

Variables	(1) # of Projects	(2) # of Units	(3) Construction area (10k m^2)
LinesCrossed $\times$ Post	-0.295*** (-4.29)	-0.432*** (-4.59)	-0.412*** (-2.99)
Size	0.314* (1.73)	0.492* (1.87)	-0.268 (-0.82)
Observations	3,864	3,864	3,864
Pseudo R^2	0.753	0.861	0.790
Firm FEs	YES	YES	YES
Year-Quarter FEs	YES	YES	YES
Cluster	Firm	Firm	Firm

Panel B: Differential effects across firm types

Variables	(1) # of Projects	(2) # of Units	(3) Construction area (10k m^2)
Private $\times$ LinesCrossed $\times$ Post	-0.241** (-2.02)	-0.407*** (-3.22)	-0.488*** (-3.12)
LinesCrossed $\times$ Post	-0.108 (-1.19)	-0.133 (-1.45)	-0.097 (-1.15)
Private $\times$ Post	-0.363** (-2.36)	-0.234** (-2.23)	0.055 (0.24)
Size	0.360** (2.18)	0.468** (2.17)	-0.350 (-0.92)
Observations	3,864	3,864	3,864
Pseudo R^2	0.759	0.867	0.795
Firm FEs	YES	YES	YES
Year-Quarter FEs	YES	YES	YES
Cluster	Firm	Firm	Firm

This table reports the Poisson regression results on the impact of the TRL policy on the introduction of new housing projects of public real estate firms covered by the CREIS Database. The data is at the firm-year-quarter level. The sample period is from 2017 to 2023. Variable definitions are provided in Appendix Table A1. Standard errors are clustered at the firm level, and z -statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6: The Impact of Three-Red-Lines Policy on Housing Demand

Panel A: Post-treatment effects

Variables	(1) Ln(Sales units)	(2) Ln(Sales area (1k m^2))	(3) Ln(Sales value (1m RMB))	(4) Ln(Sales price (RMB/ m^2))
LinesCrossed $\times$ Post	0.002 (0.05)	0.003 (0.10)	0.002 (0.06)	-0.000 (-0.16)
Size	-0.366*** (-3.60)	-0.343*** (-3.36)	-0.357*** (-3.37)	-0.012 (-1.31)
Observations	181,708	181,709	181,377	181,377
Adjusted R^2	0.497	0.527	0.539	0.953
Project FEs	YES	YES	YES	YES
City-Year-Quarter FEs	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm

Panel B: Differential effects across firm types

Variables	(1) Ln(Sales units)	(2) Ln(Sales area (1k m^2))	(3) Ln(Sales value (1m RMB))	(4) Ln(Sales price (RMB/ m^2))
Private $\times$ LinesCrossed $\times$ Post	-0.135*** (-2.76)	-0.130*** (-2.76)	-0.128*** (-2.63)	0.000 (0.00)
LinesCrossed $\times$ Post	0.088*** (3.22)	0.088*** (3.24)	0.086*** (3.01)	0.001 (0.33)
Private $\times$ Post	-0.014 (-0.12)	-0.027 (-0.25)	-0.045 (-0.40)	-0.014 (-1.45)
Size	-0.389*** (-4.06)	-0.367*** (-3.84)	-0.382*** (-3.90)	-0.013 (-1.59)
Observations	181,708	181,709	181,377	181,377
Adjusted R^2	0.497	0.528	0.540	0.953
Project FEs	YES	YES	YES	YES
City-Year-Quarter FEs	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm

This table reports the OLS regression results on the impact of the TRL policy on housing sales of public real estate firms covered by the CREIS Database. The data is at the project-year-quarter level. The sample period is from 2017 to 2023. Variable definitions are provided in Appendix Table A1. Standard errors are clustered at the firm level, and t -statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7: The TRL Policy, LGFV Propping-Up Activities, and Financial Conditions

Panel A: The impact of TRL policy on LGFV propping-up activities

Variables	(1) Propping ratio (area)	(2) Propping ratio (value)
Intensity $\times$ Post	0.295*** (4.84)	0.300*** (4.89)
Size	0.025 (1.05)	0.026 (1.06)
Observations	7,139	7,139
LGFV FEs	YES	YES
Year FEs	YES	YES
Cluster	LGFV	LGFV

Panel B: The impact of TRL policy on LGFV financial conditions

Variables	(1) LGFV yield spread	(2) Current ratio	(3) Quick ratio	(4) Cash ratio
Intensity $\times$ Post	1.475*** (4.33)	-1.861*** (-5.28)	-0.928*** (-5.34)	-0.324*** (-5.33)
Size	-0.487*** (-4.90)	-0.383** (-2.28)	-0.339*** (-4.60)	-0.099*** (-3.54)
Observations	7,139	7,139	7,139	7,139
LGFV FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES
Cluster	LGFV	LGFV	LGFV	LGFV

This table reports the OLS regression results on the impact of the TRL policy on LGFV propping-up activities, yield spreads, and financial ratios. The sample period is from 2017 to 2023. Variable definitions are provided in Appendix Table A1. Standard errors are clustered at the LGFV level, and t -statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Internet Appendix for

Deleverging Developers in China: Real Effects and Spillovers

Internet Appendix Figures

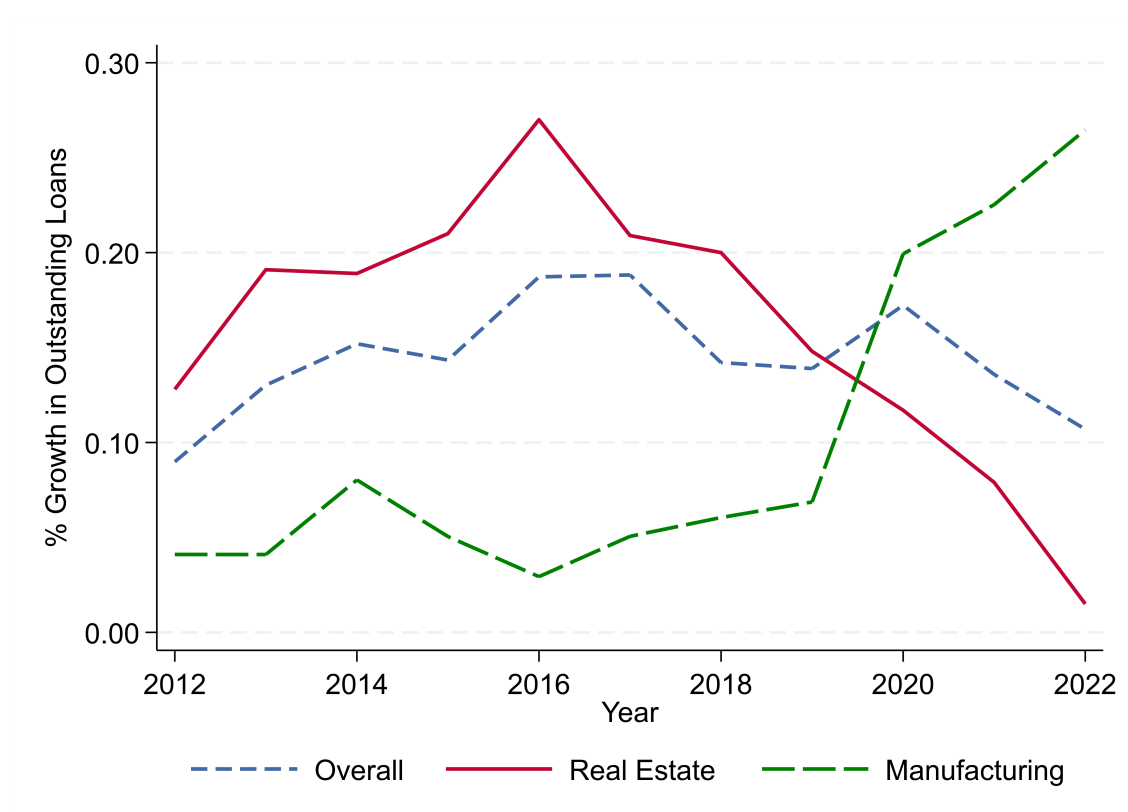


Figure IA1: Year-on-Year RMB Loan Growth: Total, Manufacturing, and Real Estate Sectors

This graph illustrates the percentage growth in outstanding RMB loans in China from 2012 to 2022. The short-dashed blue line represents the year-over-year growth in overall outstanding medium- and long-term loans. The solid red line indicates the year-over-year growth in outstanding loans to the real estate sector. The long-dashed green line shows the year-over-year growth in outstanding loans to the manufacturing sector.

Internet Appendix Tables

Table IA1: Comparing Coverage by CRIC and Land China

Year	Number of land sites			Land area (in 10k m^2)			Land value (in 10k RMB)		
	Land China	CRIC	Coverage	Land China	CRIC	Coverage	Land China	CRIC	Coverage
2013	56,832	46,976	83%	166,077	142,733	86%	380,656,480	362,558,016	95%
2014	45,153	38,852	86%	123,571	102,355	83%	293,758,752	273,943,712	93%
2015	36,639	31,506	86%	97,370	80,590	83%	265,523,392	255,781,584	96%
2016	33,847	29,971	89%	89,229	76,287	85%	311,789,600	328,380,480	105%
2017	34,411	34,563	100%	97,117	94,496	97%	438,838,112	486,636,320	111%
2018	41,693	39,726	95%	126,769	116,610	92%	544,808,832	551,591,488	101%
2019	44,023	38,054	86%	130,120	118,767	91%	625,916,608	628,685,056	100%
2020	48,300	41,579	86%	143,434	131,207	91%	745,398,912	738,198,656	99%
2021	42,561	39,664	93%	120,292	113,634	94%	699,714,624	711,241,024	102%
2022	35,837	34,242	96%	93,199	88,389	95%	514,499,424	514,375,712	100%
2023	29,954	28,435	95%	74,051	70,779	96%	422,068,096	428,447,744	102%

This table compares the aggregate coverage of land transactions recorded by China Real Estate Information Corporation (CRIC) and Land China (landchina.com). Land China, the official land management and market information platform under China's Ministry of Natural Resources, is responsible for publishing nationwide land supply plans, land use rights transfers, and transaction results. CRIC is a leading provider of real estate information and analysis. For this comparison, only land parcels acquired by firms (excluding individual acquisitions) are included, and the sample is restricted to transactions involving land designated for commercial/residential, commercial/office, and residential use.

Table IA2: Summary Statistics of Firms Crossing Different Numbers of Red Lines

Number of red lines crossed	Private				SOE			
	Number of firms	Liability-to- asset ratio	Debt-to- equity ratio	Cash-to-short term debt ratio	Number of firms	Liability-to- asset ratio	Debt-to- equity ratio	Cash-to-short term debt ratio
3	24	0.80	2.25	0.53	13	0.77	2.17	0.59
2	13	0.76	1.14	0.94	7	0.76	1.56	1.34
1	32	0.65	0.58	1.36	15	0.68	0.66	1.24
0	29	0.42	0.12	28.51	27	0.54	0.23	41.07

This table compares the summary statistics of firms crossing different numbers of red lines in 2019. Firms are further classified into private-owned firms and state-owned firms.

Table IA3: The Impact of Three-Red-Lines Policy on Housing Demand with Block FEs

Panel A: Post-treatment effects

Variables	(1) Ln(Sales units)	(2) Ln(Sales area (1k m^2))	(3) Ln(Sales value (1m RMB))	(4) Ln(Sales price (RMB/ m^2))
LinesCrossed $\times$ Post	0.012 (0.45)	0.017 (0.64)	0.016 (0.59)	-0.001 (-0.67)
Size	-0.219** (-2.37)	-0.195** (-2.08)	-0.208** (-2.15)	-0.011 (-1.31)
Observations	154,911	154,911	154,622	154,622
Adjusted R-squared	0.577	0.604	0.613	0.957
Project FEs	YES	YES	YES	YES
Block-Year-Quarter FEs	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm

Panel B: Differential effects across firm types

Variables	(1) Ln(Sales units)	(2) Ln(Sales area (1k m^2))	(3) Ln(Sales value (1m RMB))	(4) Ln(Sales price (RMB/ m^2))
Private $\times$ LinesCrossed $\times$ Post	-0.084* (-1.92)	-0.083* (-1.93)	-0.081* (-1.88)	0.001 (0.27)
LinesCrossed $\times$ Post	0.070*** (3.46)	0.075*** (3.71)	0.073*** (3.61)	-0.001 (-0.95)
Private $\times$ Post	-0.069 (-0.75)	-0.074 (-0.84)	-0.082 (-0.90)	-0.006 (-0.79)
Size	-0.246*** (-2.75)	-0.222** (-2.47)	-0.235** (-2.56)	-0.011 (-1.41)
Observations	154,911	154,911	154,622	154,622
Adjusted R-squared	0.578	0.604	0.614	0.957
Project FEs	YES	YES	YES	YES
Block-Year-Quarter FEs	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm

This table reports the OLS regression results on the impact of the TRL policy on housing sales of public real estate firms covered by the CREIS Database. The data is at the project-year-quarter level. The sample period is from 2017 to 2023. $Ln(Sales\ units)$ is the natural logarithm of the number of units sold by each project in each quarter. $Ln(Sales\ area\ (1k\ m^2))$ is the natural logarithm of the total area sold by each project in each quarter. $Ln(Sales\ value\ (1m\ RMB))$ is the natural logarithm of the total land value sold by each project in each quarter. $Ln(Sales\ price\ (RMB/m^2))$ is the natural logarithm of the average sales price across all units sold by each project in each quarter. *LinesCrossed* is a count variable that equals the number of red lines a firm crossed at the end of 2019. *Post* is a dummy variable that equals one for years on and after 2021, and zero otherwise. *Size* is the natural logarithm of a firm's total assets. *Private* is a dummy variable that equals one if a firm is a privately-owned enterprise, and zero otherwise. Standard errors are clustered at the firm level, and *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table IA4: The Impact of LGFV Propping-up Activities on LGFV Financial Conditions

<i>Panel A: Propping ratio measured by area</i>				
Variables	(1) LGFV yield spread	(2) Current ratio	(3) Quick ratio	(4) Cash ratio
Propping ratio (area)	0.078** (2.17)	-0.126*** (-2.72)	-0.093*** (-4.56)	-0.025*** (-3.36)
Size	-0.504*** (-5.03)	-0.361** (-2.13)	-0.328*** (-4.37)	-0.095*** (-3.35)
Observations	7,139	7,139	7,139	7,139
LGFV FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES
Cluster	LGFV	LGFV	LGFV	LGFV

<i>Panel B: Propping ratio measured by value</i>				
Variables	(1) LGFV yield spread	(2) Current ratio	(3) Quick ratio	(4) Cash ratio
Propping ratio (value)	0.079** (2.22)	-0.125*** (-2.73)	-0.095*** (-4.69)	-0.026*** (-3.49)
Size	-0.504*** (-5.03)	-0.361** (-2.13)	-0.328*** (-4.37)	-0.095*** (-3.35)
Observations	7,139	7,139	7,139	7,139
LGFV FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES
Cluster	LGFV	LGFV	LGFV	LGFV

This table reports the regression results on the relationship between LGFV propping-up activities and LGFV financial conditions. The dependent variables are LGFV average yield spread, current ratio, quick ratio, and cash ratio. Panel A uses *Propping ratio (area)*, measured by area. Panel B uses *Propping ratio (value)*, measured by value. Both variables are defined as the share of residential land purchased by LGFVs or their controlled subsidiaries that remains undeveloped through 2025. *Size* is the natural logarithm of an LGFV's total assets. LGFV fixed effects and year fixed effects are included in all specifications. Standard errors are clustered at the LGFV level, and *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.